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(54) **ELECTROLESS PLATING METHOD**

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(71) Applicant: **TXC Corporation**, Taipei City (TW)

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(72) Inventors: **I-Hsuan Lin**, Taipei City (TW);
Chih-Hung Chiu, Taipei City (TW)

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(73) Assignee: **TXC Corporation**, Taipei City (TW)

(57) **ABSTRACT**

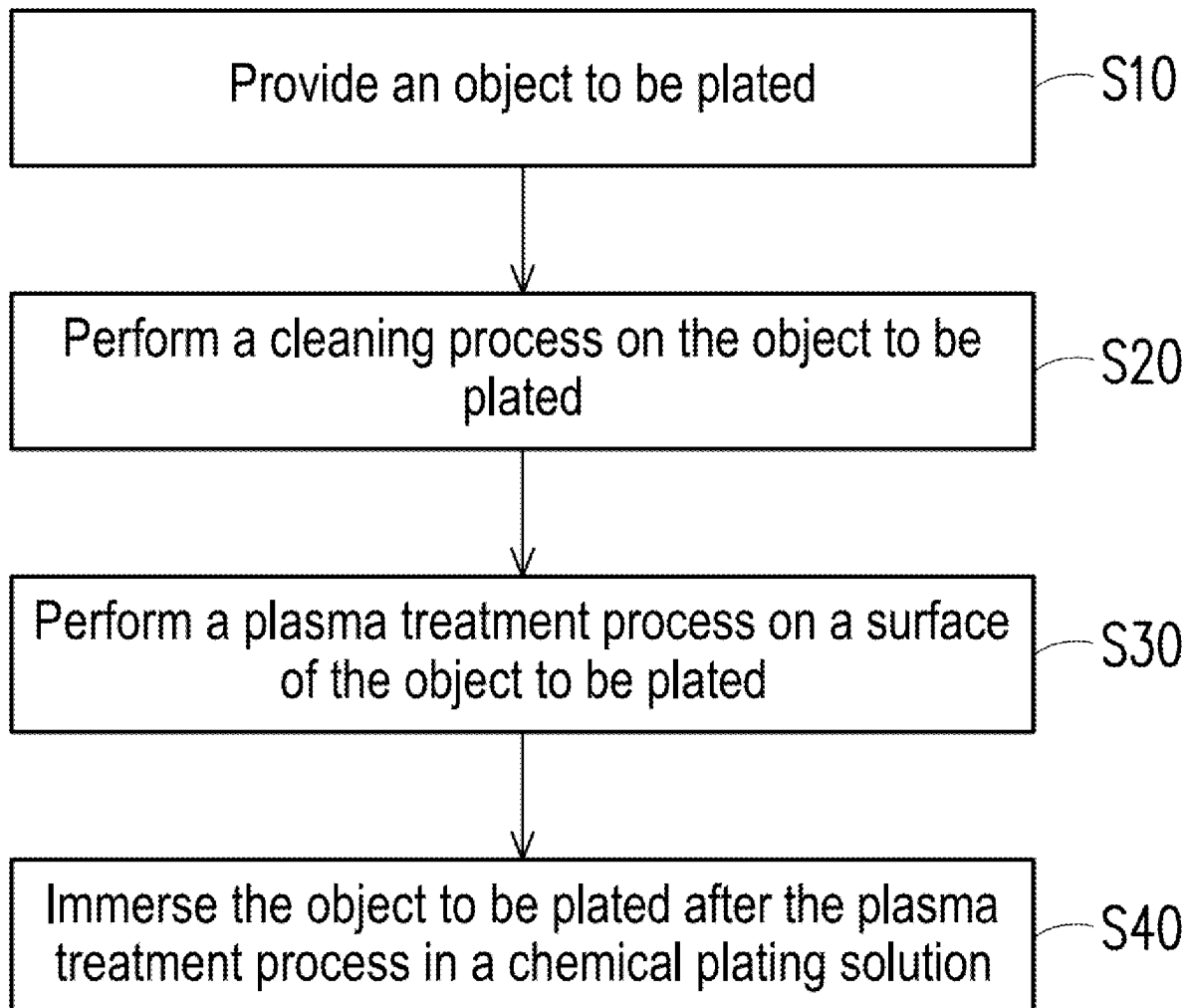
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An electroless plating method includes the following steps. An object to be plated is provided, wherein the object to be plated is a metal material. A cleaning process is performed on the object to be plated to remove impurities on a surface of the object to be plated. A plasma treatment process is performed on the surface of the object to be plated to ionize a metal of the surface of the object to be plated. The object to be plated after the plasma treatment process is immersed in a chemical plating solution to form a plating layer.

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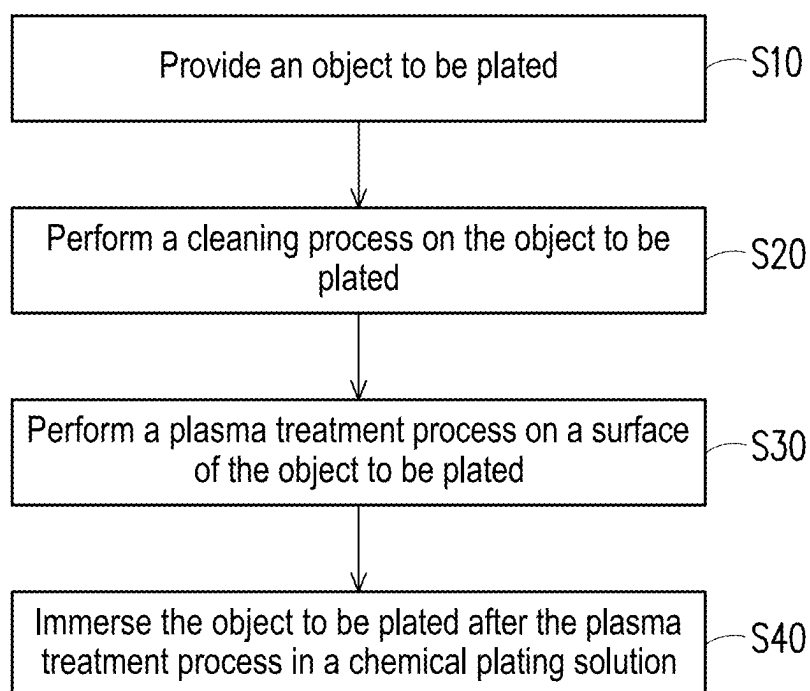


FIG. 1

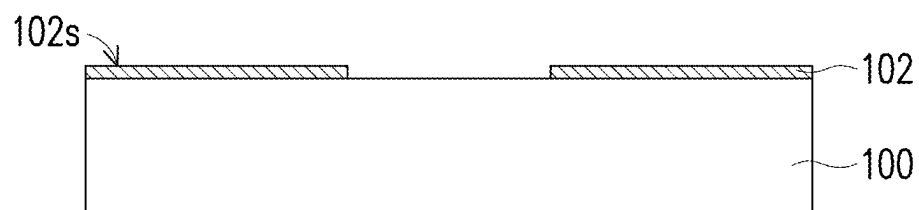


FIG. 2A

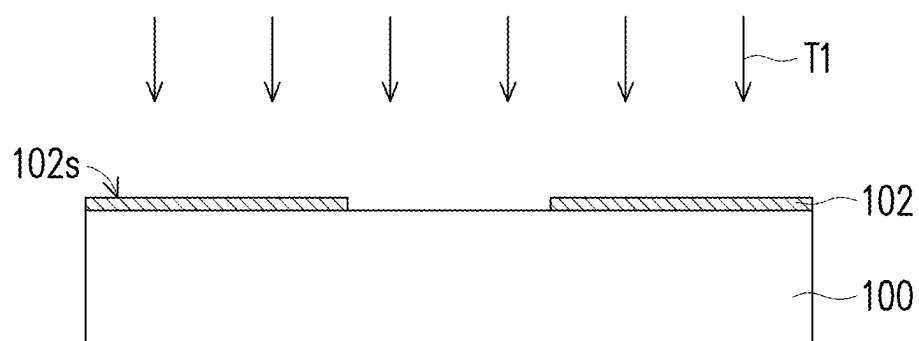


FIG. 2B

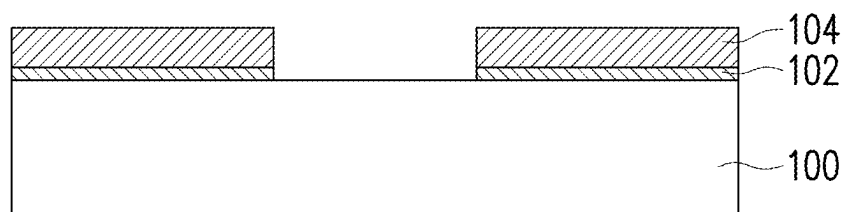


FIG. 2C

ELECTROLESS PLATING METHOD

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application serial no. 113106430, filed on Feb. 22, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of specification.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The invention relates to a plating method, and in particular to an electroless plating method.

Description of Related Art

[0003] In general, before electroless plating, sensitization and activation treatments need to be performed on the surface of the object to be plated so that the catalyst may adhere to the surface of the object to be plated, so that when the object to be plated is subsequently immersed in the chemical plating solution for reaction, precipitation and deposition of the metal on the catalyst layer may be facilitated to form a plating layer. However, sensitization and activation treatments require the selection of a corresponding chemical solution for the metal to be plated to pre-treat the object to be plated. The overall process is long and costly.

SUMMARY OF THE INVENTION

[0004] The invention provides an electroless plating method that simplify the manufacturing process and reduces manufacturing costs.

[0005] An electroless plating method of the invention includes the following steps. An object to be plated is provided, wherein the object to be plated is a metal material. A cleaning process is performed on the object to be plated to remove impurities on a surface of the object to be plated. A plasma treatment process is performed on the surface of the object to be plated to ionize a metal of the surface of the object to be plated. The object to be plated after the plasma treatment process is immersed in a chemical plating solution to form a plating layer.

[0006] In an embodiment of the invention, after the plasma treatment process of the object to be plated, the object to be plated is immersed in the chemical plating solution.

[0007] In an embodiment of the invention, a process gas in the plasma treatment process includes an oxygen gas and/or an argon gas, so that the surface of the object to be plated is exposed to a plasma environment generated by the process gas.

[0008] In an embodiment of the invention, before the object to be plated is immersed in the chemical plating solution, the surface of the object to be plated is not activated by a catalyst.

[0009] In an embodiment of the invention, a material of the object to be plated includes gold, silver, platinum, palladium, rhodium, nickel, cobalt, or an alloy thereof.

[0010] In an embodiment of the invention, the chemical plating solution includes a

[0011] metal salt, and the metal salt includes a nickel ion, a chromium ion, a gold ion, a tin ion, or a copper ion.

[0012] In an embodiment of the invention, the cleaning process includes a wet chemical cleaning method or a plasma cleaning method.

[0013] In an embodiment of the invention, a cleaning solution used in the wet chemical cleaning method includes hydrochloric acid, sulfuric acid, hydrogen peroxide, ammonia, dilute hydrofluoric acid, organic acid, sodium hydroxide, or a combination thereof.

[0014] In an embodiment of the invention, the cleaning process is a plasma cleaning method, and the cleaning process and the plasma treatment process are performed in a same process step.

[0015] In an embodiment of the invention, the object to be plated is formed on a substrate, and a material of the substrate includes a quartz, a glass, a resin, or a semiconductor.

[0016] Based on the above, the invention may significantly simplify the process steps and reduce the manufacturing cost by replacing the surface catalyst activation treatment with the surface plasma treatment process.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a flowchart of an electroless plating method according to an embodiment of the invention.

[0018] FIG. 2A to FIG. 2C are schematic cross-sectional views of various stages of an electroless plating method according to an embodiment of the invention.

DESCRIPTION OF THE EMBODIMENTS

[0019] The directional terms mentioned herein, such as “upper”, “lower”, “front”, “rear”, “left”, “right”, etc., refer to the directions of the drawings. Accordingly, the directional terms used are illustrative, not limiting, of the invention.

[0020] In the drawings, each figure illustrates the general characteristics of methods, structures, and/or materials used in particular embodiments. However, these drawings should not be interpreted as defining or limiting the scope or nature encompassed by these embodiments. For example, the relative dimensions, thicknesses, and positions of layers, regions, and/or structures may be reduced or exaggerated for clarity.

[0021] In the following embodiments, the same or similar elements are given the same or similar reference numerals, and descriptions thereof are omitted. Moreover, features in different embodiments may be combined with each other without conflict, and simple equivalent changes and modifications made in accordance with this specification or the claims are still within the scope of this patent.

[0022] FIG. 1 is a flowchart of an electroless plating method according to an embodiment of the invention. FIG. 2A to FIG. 2C are schematic cross-sectional views of various stages of an electroless plating method according to an embodiment of the invention.

[0023] Please refer to FIG. 1 and FIG. 2A. In step S10, an object 102 to be plated is provided. The object 102 to be plated may be a metal material. For example, the material of the object 102 to be plated may include gold, silver, platinum, palladium, rhodium, nickel, cobalt, an alloy thereof, or

other suitable metal materials to make subsequent chemical plating readily occur on a surface 102s of the object 102 to be plated.

[0024] In some embodiments, the object 102 to be plated may be formed on the substrate 100. For example, the object 102 to be plated may be formed on the substrate 100 via physical vapor deposition, chemical vapor deposition, sputtering, electroplating, or other deposition methods. The material of the substrate 100 may include a quartz, a glass, a resin, a semiconductor, or other suitable substrate materials, but the invention is not limited thereto. The detailed structure of the substrate 100 is omitted in FIG. 2A, but it should be understood that the substrate 100 may include a structure such as a conductive layer, a conductive via, so that the object 102 to be plated may be electrically connected to the corresponding conductive structure in the substrate 100.

[0025] Please continue to refer to FIG. 1 and FIG. 2A. In step S20, a cleaning process is performed on the object to be plated 102 to remove impurities (such as oxides, organic matter, metal particles, etc.) on the surface 102s of the object to be plated 102. For example, the cleaning process may include wet chemical cleaning, plasma cleaning, or other suitable cleaning methods. The wet chemical cleaning method removes impurities using a chemical reaction between the cleaning solution and the impurities on the surface 102s of the object 102 to be plated. The cleaning solution may include hydrochloric acid, sulfuric acid, hydrogen peroxide, ammonia, dilute hydrofluoric acid, organic acid, sodium hydroxide, a combination thereof, or other suitable cleaning solutions. In some embodiments, the cleaning solution includes hydrogen peroxide and ammonia to remove organic contaminants or particles at the surface 102s of the object 102 to be plated. In some embodiments, the cleaning solution includes hydrogen peroxide and hydrochloric acid to remove metal or ionic contaminants at the surface 102s of the object 102 to be plated. In some embodiments, the cleaning solution includes hydrogen peroxide and sulfuric acid to remove organic contaminants at the surface 102s of the object 102 to be plated. In some embodiments, the cleaning solution includes dilute hydrofluoric acid to remove oxides from the surface 102s of the object 102 to be plated. In some embodiments, the wet chemical cleaning method may perform cleaning multiple times using different cleaning solutions, but the invention is not limited thereto. The plasma cleaning method removes impurities using plasma to chemically react with impurities at the surface 102s of the object 102 to be plated or using plasma to impact the surface 102s of the object 102. In some embodiments, the plasma cleaning method is performed in a plasma environment generated by a gas source containing oxygen gas, hydrogen gas, argon gas, a combination thereof, or other suitable gases. In some embodiments, the object to be plated 102 may be cleaned via one or more wet chemical cleaning methods, one or more plasma cleaning methods, or a combination of wet chemical cleaning methods and plasma cleaning methods, but the invention is not limited thereto.

[0026] Please refer to FIG. 1 and FIG. 2B. In step S30, a plasma treatment process T1 is performed on the surface 102s of the object 102 to be plated to ionize the metal of the surface 102s of the object 102 to be plated. For example, the process gas of the plasma treatment process T1 may include oxygen gas, argon gas, a combination thereof, and/or other suitable process gases, so that the surface 102s of the object to be plated 102 is exposed to the plasma environment

generated by the process gas. Therefore, the metal of the surface thereof is ionized, so that the surface energy and hydrophilicity are improved to facilitate the forming of the subsequent plating layer thereon without the need to perform sensitization and activation treatments on the surface 102s of the object 102 to be plated, thus simplifying the process steps.

[0027] In some embodiments, when the cleaning process is a plasma cleaning method, the cleaning process and the plasma treatment process may be combined and performed in the same process step, but the invention is not limited thereto. In other embodiments, the cleaning process and the plasma treatment process may be performed in different process steps.

[0028] Please refer to FIG. 1 and FIG. 3B, in step S40, the object to be plated 102 after the plasma treatment process T1 is immersed in a chemical plating solution (not shown) to generate a chemical reaction at the surface 102s of the object to be plated 102 to form a plating layer 104. In some embodiments, the chemical plating solution includes metal salt to provide a metal ion source of the plating layer 104. For example, the metal salt includes nickel ion, chromium ion, gold ion, tin ion, copper ion, or other suitable metal ions, depending on the material of the plating layer 104 to be plated. In some embodiments, the metal salt is, for example, a salt formed by metal ion and sulfate, sulfite, chloride ion, or the like, but the invention is not limited thereto. In some embodiments, the chemical plating solution further includes a reducing agent to reduce the metal ion into metal precipitation. For example, the reducing agent may be selected from the group consisting of sodium hypophosphite, sodium borohydride, amine boron, formic acid, formaldehyde, sodium formate, and hydroxylamine. In some embodiments, the chemical plating solution further includes a complexing agent and a buffer to stabilize the metal ion in the chemical plating solution. In some embodiments, the chemical plating solution may be heated to reach the working temperature to facilitate the chemical reaction.

[0029] In some embodiments, after the plasma treatment process T1 of the object to be plated 102, the object to be plated 102 is immersed in the chemical plating solution to facilitate the chemical reaction on the surface 102s of the object to be plated 102, so that the plating layer 104 may be uniformly formed on the surface 102s of the object 102 to be plated.

[0030] In the present embodiment, before the object 102 to be plated is immersed in the chemical plating solution, the surface 102s of the object 102 to be plated is not activated by a catalyst. That is, before the object 102 to be plated is immersed in the chemical plating solution, no catalyst layer is formed at the surface 102s of the object 102 to be plated. Since the surface 102s of the object 102 to be plated of the present embodiment does not need to be activated by a catalyst, complicated surface sensitization and activation treatment steps may be omitted, thereby reducing manufacturing costs.

[0031] In some embodiments, the object 102 to be plated may be regarded as a seed layer of the plating layer 104. In some embodiments, the object 102 to be plated and the plating layer 104 may be regarded as pads of the substrate 100, so that the substrate 100 may be connected to the outside via the pads. In other embodiments, the object 102 to be plated and the plating layer 104 may serve as barrier layers to provide protection for a subsequent process.

[0032] Based on the above, the invention may significantly simplify the process steps and reduce the manufacturing cost by replacing the surface sensitization and activation treatments with a surface plasma treatment process.

[0033] It will be apparent to those skilled in the art that various modifications and variations may be made to the structure of the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims and their equivalents.

What is claimed is:

1. An electroless plating method, comprising:
providing an object to be plated, wherein the object to be plated is a metal material;
performing a cleaning process on the object to be plated to remove impurities on a surface of the object to be plated;
performing a plasma treatment process on the surface of the object to be plated to ionize a metal of the surface of the object to be plated; and
immersing the object to be plated after the plasma treatment process in a chemical plating solution to form a plating layer.
2. The electroless plating method of claim 1, wherein after the plasma treatment process of the object to be plated, the object to be plated is immersed in the chemical plating solution.
3. The electroless plating method of claim 1, wherein a process gas of the plasma treatment process comprises an oxygen gas and/or an argon gas, so that the surface of the

object to be plated is exposed to a plasma environment generated by the process gas.

4. The electroless plating method of claim 1, wherein before the object to be plated is immersed in the chemical plating solution, the surface of the object to be plated is not activated by a catalyst.

5. The electroless plating method of claim 1, wherein a material of the object to be plated comprises gold, silver, platinum, palladium, rhodium, nickel, cobalt, or an alloy thereof.

6. The electroless plating method of claim 1, wherein the chemical plating solution comprises a metal salt, and the metal salt comprises a nickel ion, a chromium ion, a gold ion, a tin ion, or a copper ion.

7. The electroless plating method of claim 1, wherein the cleaning process comprises a wet chemical cleaning method or a plasma cleaning method.

8. The electroless plating method of claim 7, wherein a cleaning solution used in the wet chemical cleaning method comprises hydrochloric acid, sulfuric acid, hydrogen peroxide, ammonia, dilute hydrofluoric acid, organic acid, sodium hydroxide, or a combination thereof.

9. The electroless plating method of claim 7, wherein the cleaning process is a plasma cleaning method, and the cleaning process and the plasma treatment process are performed in a same process step.

10. The electroless plating method of claim 1, wherein the object to be plated is formed on a substrate, and a material of the substrate comprises a quartz, a glass, a resin, or a semiconductor.

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